

Exercise class 2

1.) Prove using Parseval's identity

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Recall Parseval's identity:

$$2a_0^2 + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)^2 dx$$

Recall from the first class:

$$f(x) = x \quad -\pi \leq x < \pi \quad \text{and} \quad f(x+\pi) = f(x)$$

$$f(x) = \sum_{n=1}^{\infty} -\frac{2}{n} (-1)^n \sin(nx)$$

$$a_n = 0 \quad n = 0, 1, 2, \dots$$

$$b_n = -\frac{2}{n} (-1)^n$$

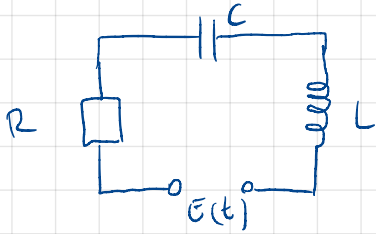
$$\text{Parseval:} \quad \sum_{n=1}^{\infty} \left[-\frac{2}{n} (-1)^n \right]^2 = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{1}{\pi} \left[\frac{x^3}{3} \right]_{-\pi}^{\pi} = \frac{2\pi^2}{3}$$

$$\hookrightarrow \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{2\pi^2}{3} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \quad \square$$

Forced oscillations

$$m \cdot \ddot{y} = \underset{\substack{\uparrow \\ \text{spring force}}}{-k \cdot y} - \underset{\substack{\uparrow \\ \text{damping}}}{C \cdot \dot{y}} + \underset{\substack{\uparrow \\ \text{driving}}}{r(t)}$$

Analogy system: RLC circuit:

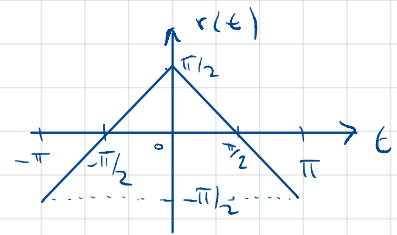


$$L \cdot \ddot{I} + R \dot{I} + \frac{1}{C} I = \dot{E}(t)$$

$$m = 1 \text{ g}, \quad C = 0,05 \text{ s}^2/\text{s}, \quad k = 25 \text{ g/s}^2$$

$$\ddot{y} + 0,05 \dot{y} + 25 y = r(t)$$

$$r(t) = \begin{cases} t + \pi/2 & -\pi < t < 0 \\ -t + \pi/2 & 0 < t < \pi \end{cases}$$



$$r(t) = a_0 + \sum_n \left[a_n \cos(n t) + b_n \sin(n t) \right]$$

$$b_n = 0, \quad a_0 = 0, \quad a_n \stackrel{n \text{ odd}}{\downarrow} = 0, \quad a_n \stackrel{n \text{ even}}{\downarrow} = \frac{4}{\pi} \frac{1}{n^2} \quad \text{/see below/}$$

One term: $\ddot{y} + 0,05 \dot{y} + 25 y = \frac{4}{n^2 \pi} \cos(n t)$

$$y_n = A_n \cos(n t) + B_n \sin(n t)$$

$$\begin{aligned} & -A_n n^2 \cos(n t) - B_n n^2 \sin(n t) + 0,05 (n A_n \sin(n t) + B_n n \cos(n t)) \\ & + 25 (A_n \cos(n t) + B_n \sin(n t)) = \frac{4}{n^2 \pi} \cos(n t) \end{aligned}$$

$$\left. \begin{aligned} \cos(n t): & \quad -A_n n^2 + 0,05 n B_n + 25 A_n = \frac{4}{n^2 \pi} \\ \sin(n t): & \quad -B_n n^2 - 0,05 n A_n + 25 B_n = 0 \end{aligned} \right\}$$

$$A_n = \frac{25-n^2}{0,054} B_n = 20 \frac{25-n^2}{n} B_n$$

$$B_n \left[\frac{20}{n} (25-n^2)^2 + n/20 \right] = \frac{4}{n^2} \pi$$

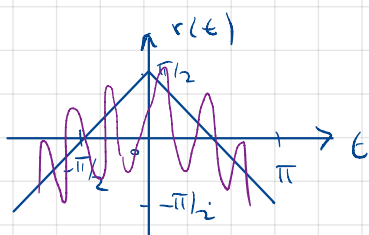
$$B_n = \frac{4}{\pi n} \frac{1}{20 (25-n^2)^2 + n^2/20} \rightarrow A_n = \frac{80}{\pi n^2} \frac{25-n^2}{20(25-n^2)^2 + n^2/20}$$

amplitude: $C_n = \sqrt{A_n^2 + B_n^2}$

$$C_1 = 0,0531, \quad C_3 = 0,0088, \quad C_5 = 0,2037, \quad C_7 = 0,0011, \quad \dots$$

↑
dominates!

without damping and driving,
this is the oscillator frequency



$$a_0 = 0, \quad b_0 = 0$$

$$a_n > 0 = \frac{1}{\pi} \int_{-\pi}^{\pi} r(t) \cos(nt) dt = \frac{2}{\pi} \int_0^{\pi} \left(\frac{\pi}{2} - t \right) \cos(nt) dt$$

$$= \frac{2}{\pi} \cdot \frac{\pi}{2} \left[\frac{1}{n} \sin(nt) \right]_0^{\pi} - \frac{2}{\pi} \int_0^{\pi} t \left(\frac{1}{n} \sin(nt) \right)' dt$$

$$= 0 - \frac{2}{\pi} \left[\frac{t}{n} \sin(nt) \right]_0^{\pi} + \frac{2}{\pi} \int_0^{\pi} \frac{1}{n} \sin(nt) dt$$

$$= \frac{2}{\pi} \frac{1}{n} \left[-\frac{1}{n} \cos(nt) \right]_0^{\pi} = -\frac{2}{\pi n^2} \left((-1)^n - 1 \right) = \begin{cases} \frac{4}{\pi n^2} & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$$

Fourier coeffs.:

Fourier sine/cosine transform

Different convention from book!

$$f(x) = \int_0^\infty d\omega [A(\omega) \cos(\omega x) + B(\omega) \sin(\omega x)]$$

where $A(\omega) = \frac{1}{\pi} \int_{-\infty}^\infty dx \cos(\omega x) f(x)$ cos transform

$$B(\omega) = \frac{1}{\pi} \int_{-\infty}^\infty dx \sin(\omega x) f(x) \quad \text{sin trf.}$$

Other notations $A(\omega) = \widehat{f_c}(\omega) = \mathcal{F}_c(f)(\omega)$

$$B(\omega) = \widehat{f_s}(\omega) = \mathcal{F}_s(f)(\omega)$$

Properties

- $\widehat{f_c}(\omega) = \widehat{f_c}(-\omega)$

proof

$$\widehat{f_c}(\omega) = \frac{1}{\pi} \int_{-\infty}^\infty dx \cos(\omega x) f(x) = \frac{1}{\pi} \int_{-\infty}^\infty dx \cos(-\omega x) f(x) = \widehat{f_c}(-\omega) \quad \text{cos even}$$

- $\widehat{f_s}(\omega) = -\widehat{f_s}(-\omega)$

proof: similar, sin odd

- If $g(x) = f(-x)$ then $\widehat{g_c}(\omega) = \widehat{f_c}(-\omega) = \widehat{f_c}(\omega)$
 $\widehat{g_s}(\omega) = \widehat{f_s}(-\omega) = -\widehat{f_s}(\omega)$

proof:

$$\begin{aligned} \widehat{g_c}(\omega) &= \frac{1}{\pi} \int_{-\infty}^\infty dx \cos(\omega x) f(-x) = \frac{1}{\pi} \int_{-\infty}^\infty (-dy) \cos(-\omega y) f(y) = \\ &= \frac{1}{\pi} \int_{-\infty}^\infty dy \cos(\omega y) f(y) = \widehat{f_c}(\omega) \end{aligned}$$

similar for $\widehat{g_s}$

• If $g(x) = f'(x)$ (derivative)

then $\hat{g}_c(\omega) = \omega \hat{f}_s(\omega)$

$$\hat{g}_s(\omega) = -\omega \hat{f}_c(\omega)$$

proof:

$$\begin{aligned} f'(x) &= \frac{d}{dx} \int_0^{\infty} d\omega \left(\hat{f}_c(\omega) \cos(\omega x) + \hat{f}_s(\omega) \sin(\omega x) \right) = \\ &= \int_0^{\infty} d\omega \left(\hat{f}_c(\omega) \omega [-\sin(\omega x)] + \hat{f}_s(\omega) \omega \cos(\omega x) \right) \end{aligned}$$

compare with

$$g(x) = \int_0^{\infty} d\omega \left(\hat{g}_c(\omega) \cos(\omega x) + \hat{g}_s(\omega) \sin(\omega x) \right)$$

$$\Rightarrow \hat{g}_c(\omega) = \omega \hat{f}_s(\omega) \quad \hat{g}_s(\omega) = -\omega \hat{f}_c(\omega)$$

• Linearity: if $h(x) = a f(x) + b g(x)$

then $\hat{h}_{c/s}(\omega) = a \hat{f}_{c/s}(\omega) + b \hat{g}_{c/s}(\omega)$

• Rescaling $g(x) = f(ax) \quad a > 0$

$$\begin{aligned} \hat{g}_c(\omega) &= \frac{1}{\pi} \int_{-\infty}^{\infty} dx \, f(ax) \cos(\omega x) = \frac{1}{\pi} \int_{-\infty}^a \frac{dy}{a} f(y) \cos\left(\frac{\omega}{a} y\right) = \\ &= \frac{1}{a} \hat{f}_c\left(\frac{\omega}{a}\right) \end{aligned}$$

Similarly $\hat{g}_s(\omega) = \frac{1}{a} \hat{f}_s\left(\frac{\omega}{a}\right)$

• Cosine transform of $f(x) = \begin{cases} e^{-x} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases} \quad (*)$

$$\hat{f}_c(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} dx f(x) \cos(\omega x) = \frac{1}{\pi} \int_0^{\infty} dx \underbrace{e^{-x}}_u \underbrace{\cos(\omega x)}_v = \quad \text{I.B.P.}$$

$$u' = -e^{-x} \quad v' = \frac{1}{\omega} \sin(\omega x)$$

$$= \frac{1}{\pi} \left[\underbrace{e^{-x} \cdot \frac{1}{\omega} \sin(\omega x)}_0 \right]_0^{\infty} + \frac{1}{\pi} \int_0^{\infty} \frac{e^{-x} \sin(\omega x)}{\omega} dx = \quad \text{I.B.P.}$$

$$u = e^{-x} \quad u' = -e^{-x}$$

$$v' = \sin(\omega x) \quad v = -\frac{1}{\omega} \cos(\omega x)$$

$$= \frac{1}{\pi} \left[-\frac{1}{\omega^2} e^{-x} \cos(\omega x) \right]_0^{\infty} - \frac{1}{\pi} \frac{1}{\omega^2} \int_0^{\infty} e^{-x} \cos(\omega x) dx =$$

$$\hat{f}_c(\omega) = \frac{1}{\pi} \frac{1}{\omega^2} - \frac{1}{\omega^2} \hat{f}_c(\omega)$$

$$\hat{f}_c(\omega) \left(1 + \frac{1}{\omega^2} \right) = \frac{1}{\pi} \frac{1}{\omega^2}$$

$$\hat{f}_c(\omega) = \frac{1}{\pi} \frac{1}{1 + \omega^2}$$

• Cos. trf. of $f(x) = \begin{cases} e^{-ax} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases} \quad a > 0$

Use rescaling $\hat{f}_c(\omega) = \frac{1}{a} \frac{1}{\pi} \frac{1}{1 + \left(\frac{\omega}{a}\right)^2} = \frac{1}{\pi} \frac{a}{\omega^2 + a^2}$

• Cos. trf of $f_2(x) = e^{-|x|}$ use f from (*)

$$f_2(x) = f(x) + f(-x)$$

$$\hat{f}_{2c}(x) = \hat{f}_c(x) + \hat{f}_c(x) = \frac{2}{\pi} \frac{1}{1 + \omega^2}$$

• We know that for $f(x) = e^{-x^2}$

$$\hat{f}_c(\omega) = \sqrt{\pi} e^{-\frac{\omega^2}{4}} \quad (\text{and } \hat{f}_s(\omega) = 0)$$

What is the F.T. of $g(x) = x e^{-x^2}$?

Solution:

$$g(x) = -\frac{1}{2} f'(x)$$

$$\hat{g}_c(\omega) = -\frac{1}{2} \omega \hat{f}_s(\omega) = 0$$

$$\hat{g}_s(\omega) = \frac{1}{2} \omega \hat{f}_c(\omega) = \frac{\sqrt{\pi}}{2} \omega e^{-\frac{\omega^2}{4}}$$